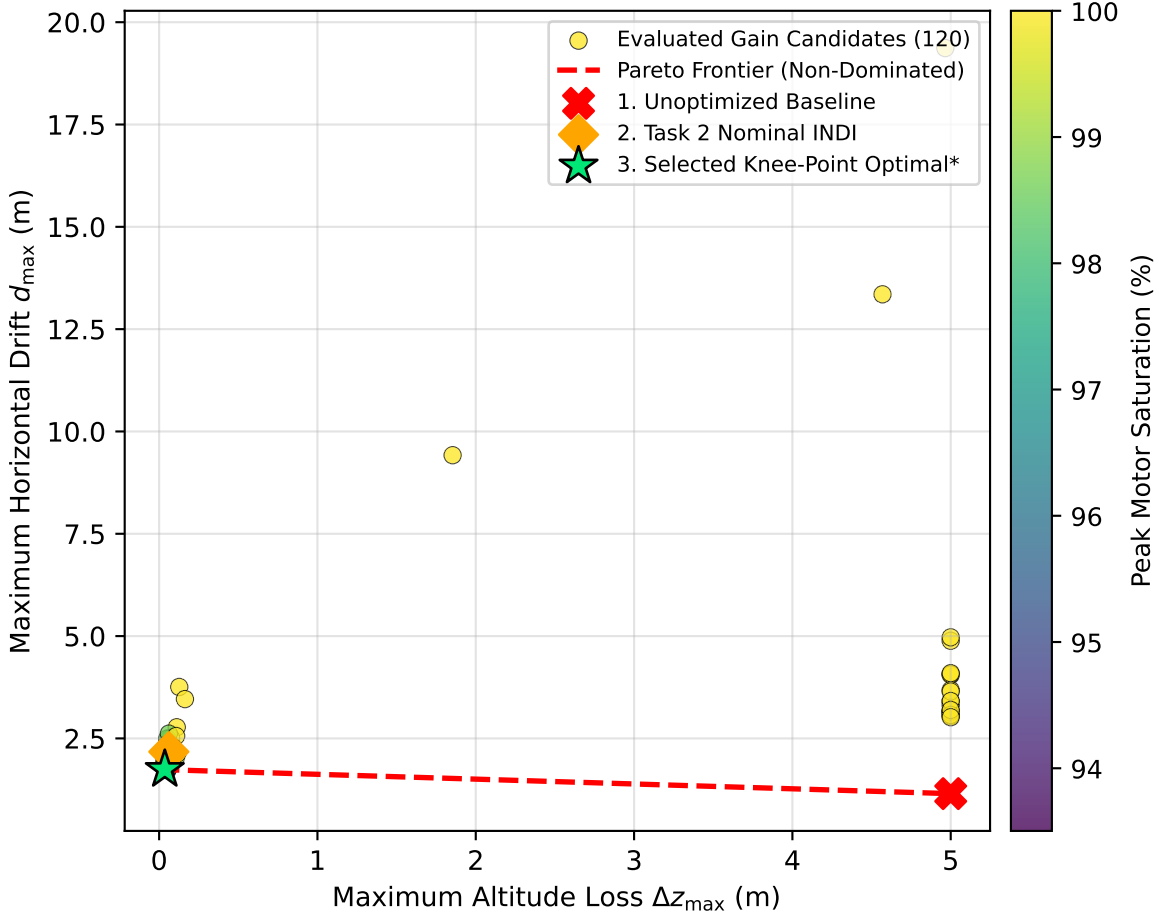
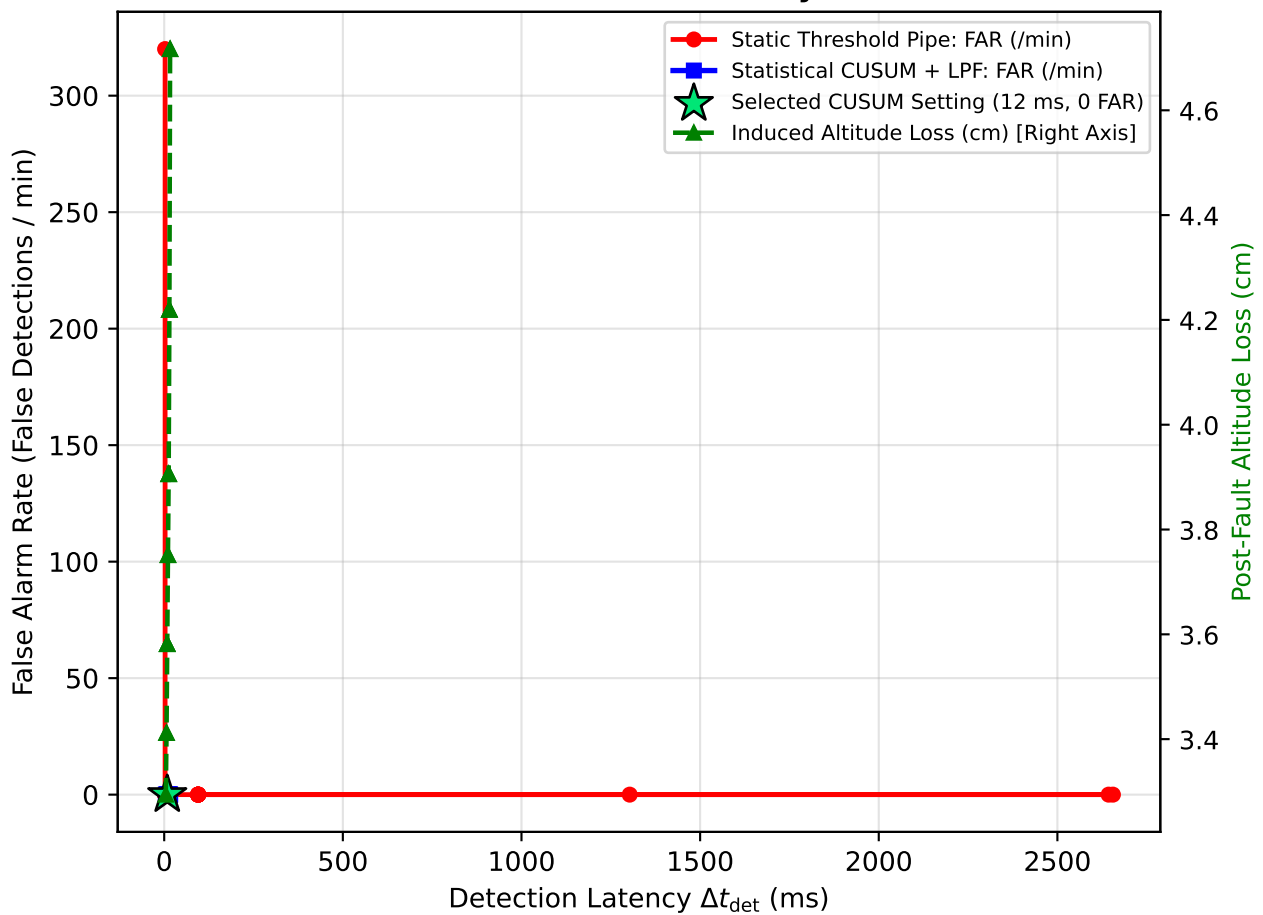

1. Controller Pareto Trade-Off: Drift vs. Altitude Loss
Evaluated Gain Candidates (120)
Pareto Frontier (Non-Dominated)
1. Unoptimized Baseline
2. Task 2 Nominal INDI
3. Selected Knee-Point Optimal*
Maximum Horizontal Drift d_{max} (m)
Maximum Altitude Loss Δz_{max} (m)
Peak Motor Saturation (%)
20.0
17.5
15.0
12.5
10.0
7.5
5.0
2.5
0
1
2
3
4
5
100
99
98
97
96
95
94


2. Detector Pareto Trade-Off: Latency vs. False Alarms
Static Threshold Pipe: FAR (/min)
Statistical CUSUM + LPF: FAR (/min)
Selected CUSUM Setting (12 ms, 0 FAR)
Induced Altitude Loss (cm) [Right Axis]
False Alarm Rate (False Detections / min)
Detection Latency Δt_{det} (ms)
Post-Fault Altitude Loss (cm)
300
250
200
150
100
50
0
0
500
1000
1500
2000
2500
4.6
4.4
4.2
4.0
3.8
3.6
3.4